

BARRA, United KingdomWMO: **030350**Lat: **57.033N**Lon: **7.450W**Elev: **3**StdP: **101.29**Time Zone: **0.00 (EUW)**Period: **97-07**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	0.8	1.7	-3.4	2.8	3.1	-2.4	3.1	3.4	22.4	5.8	21.0	6.2	5.0	10	0.835

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.1	18.9	16.2	17.6	15.3	16.6	14.6	16.8	18.2	15.9	17.0	15.2	16.1	5.7	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.2	11.5	17.6	15.4	10.9	16.6	14.7	10.5	15.8	47.2	18.3	44.6	17.0	42.6	16.2	19.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.8	16.5	14.9	-1.9	21.7	3.2	1.8	-4.2	23.0	-6.1	24.0	-7.9	25.0	-10.3	26.3		
(5)			-3.0	18.1	2.9	1.2	-5.1	19.0	-6.8	19.7	-8.4	20.4	-10.6	21.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.0	6.9	6.5	7.0	8.3	10.3	12.3	13.8	14.5	13.3	10.7	8.8	7.2
	DBStd	3.39	1.91	2.23	2.06	2.17	1.77	1.71	1.60	1.52	1.48	1.95	1.98	2.28
	HDD10.0	514	96	99	93	57	17	2	1	0	1	16	46	88
	HDD18.3	3044	354	332	350	301	248	182	141	118	152	236	287	344
	CDD10.0	512	1	1	1	7	28	70	118	141	98	38	8	2
	CDD18.3	1	0	0	0	0	0	0	1	1	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	7.4	9.1	9.2	7.9	7.1	6.1	6.2	5.4	5.7	7.0	8.1	8.7	8.7
Precipitation	PrecAvg	1304	157	114	105	78	66	72	82	101	104	138	140	139
	PrecMax	1717	261	263	216	128	162	134	141	266	205	240	230	242
	PrecMin	960	67	48	19	34	8	23	32	42	27	46	89	28
	PrecStd	173	49	53	41	24	35	27	25	46	42	56	33	55
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.6	11.7	11.9	15.5	18.2	19.5	20.7	21.0	17.7	15.1	13.6	12.5
		MCWB	10.8	10.8	10.4	11.3	14.3	15.9	17.1	18.0	15.1	13.8	12.5	11.2
	2%	DB	11.0	11.0	11.1	13.2	15.5	16.9	18.3	19.2	16.7	14.5	12.8	11.7
		MCWB	10.3	10.2	9.8	10.3	12.4	14.1	16.0	16.8	14.6	13.1	11.6	10.7
	5%	DB	10.4	10.4	10.5	12.0	13.9	15.7	17.0	17.7	16.1	14.0	12.1	11.1
		MCWB	9.6	9.5	9.3	10.1	11.8	13.5	15.1	15.7	14.4	12.6	10.8	10.2
	10%	DB	9.7	9.7	9.9	11.2	12.9	14.8	16.1	16.8	15.5	13.5	11.6	10.5
		MCWB	8.8	8.6	8.7	9.6	11.1	12.9	14.3	15.0	13.9	12.1	10.3	9.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.2	11.3	11.1	11.9	14.4	16.0	17.8	18.4	16.2	14.5	12.8	11.8
		MCDB	11.4	11.6	11.4	13.7	17.1	18.9	20.1	20.5	16.9	14.7	13.3	12.1
	2%	WB	10.6	10.5	10.4	11.0	13.0	14.7	16.6	17.1	15.4	13.8	12.2	11.1
		MCDB	10.8	10.9	10.8	12.3	14.6	16.1	17.9	18.5	16.0	14.2	12.6	11.5
	5%	WB	9.8	9.7	9.7	10.4	12.1	14.0	15.6	16.3	14.8	13.1	11.2	10.4
		MCDB	10.2	10.2	10.2	11.7	13.4	15.2	16.5	17.4	15.7	13.7	11.8	10.9
	10%	WB	9.0	8.8	9.0	9.8	11.5	13.3	14.9	15.5	14.2	12.5	10.7	9.7
		MCDB	9.6	9.5	9.6	11.0	12.6	14.4	15.8	16.5	15.1	13.3	11.4	10.4
Mean Daily Temperature Range	5% DB	MDBR	3.2	3.5	3.8	4.2	4.6	4.3	4.2	4.1	3.7	3.2	3.2	3.4
		MCDBR	3.7	3.3	3.9	4.9	5.7	5.5	5.1	5.2	4.2	3.3	3.2	3.3
		MCWBR	4.1	3.4	3.3	3.4	3.9	3.8	3.3	3.6	3.2	3.1	3.8	3.9
	5% WB	MCDBR	3.5	3.2	3.5	4.4	5.1	4.8	4.7	4.9	3.5	3.2	3.2	3.4
		MCWBR	4.0	3.6	3.4	3.4	3.8	3.8	3.3	3.7	3.1	3.7	4.1	4.2
Clear-Sky Solar Irradiance	taub	0.355	0.361	0.377	0.387	0.390	0.377	0.380	0.370	0.373	0.355	0.340	0.334	
	taud	2.132	2.238	2.285	2.316	2.366	2.445	2.449	2.483	2.434	2.363	2.209	2.116	
	Ebn at Noon	474	667	769	826	848	864	853	838	774	675	496	387	
	Edh at Noon	60	80	98	110	111	104	102	92	84	70	54	46	
All-Sky Solar Radiation	RadAvg	0.39	1.02	2.20	3.82	5.01	5.11	4.69	3.76	2.48	1.29	0.52	0.26	
	RadStd	0.04	0.10	0.16	0.26	0.37	0.51	0.32	0.25	0.13	0.13	0.05	0.03	